CBSE 2027 Chemistry — Predicted Paper (Set 10 of 10)

Based on the highly repeated questions from previous board exams (PYQs), official CBSE directives, and the most critical NCERT concepts, here is the final Set 10 of the predicted board paper. All questions are entirely fresh, uniquely formulated, and focus on heavily tested reaction mechanisms, analytical reasoning, and high-probability numericals.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. When chloroform ($CHCl_3$) and acetone (CH_3COCH_3) are mixed, hydrogen bonding occurs between them. What will be the sign of the enthalpy of mixing (ΔH_{mix}) for this solution? (a) $\Delta H_{mix} > 0$ (b) $\Delta H_{mix} < 0$ (c) $\Delta H_{mix} = 0$ (d) Cannot be determined * **Chapter:** Solutions | **Confidence:** Very High

Q2. The quantity of electricity required to completely reduce 1 mole of Cu^{2+} ions from an aqueous solution of $CuSO_4$ to Cu metal is: (a) $1F$ (b) $2F$ (c) $0.5F$ (d) $4F$ * **Chapter:** Electrochemistry | **Confidence:** High

Q3. Which of the following equations represents the integrated rate law for a zero-order reaction? (a) $[R] = -kt + [R]_0$ (b) $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$ (c) $[R] = [R]_0 e^{-kt}$ (d) $\frac{1}{[R]} - \frac{1}{[R]_0} = kt$ * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q4. What is the spin-only magnetic moment of an Fe^{2+} ion (Atomic number $Z = 26$) in an aqueous solution? (a) 1.73 BM (b) 2.83 BM (c) 3.87 BM (d) 4.90 BM * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q5. What is the primary oxidation state of Iron (Fe) in the coordination complex $K_3[Fe(CN)_6]$? (a) +2 (b) +3 (c) +4 (d) +6 * **Chapter:** Coordination Compounds | **Confidence:** High

Q6. Which of the following alkyl halides will undergo an S_N2 reaction at the fastest rate? (a) $CH_3 - Cl$ (b) $CH_3 - Br$ (c) $CH_3 - I$ (d) $CH_3 - F$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q7. The most suitable reagent for the selective oxidation of a primary alcohol to an aldehyde without further oxidation to a carboxylic acid is: (a) Acidified $K_2Cr_2O_7$ (b) Alkaline $KMnO_4$ (c) Pyridinium Chlorochromate (PCC) (d) $LiAlH_4$ * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q8. The reagent used in the Stephen reaction to reduce a nitrile to an aldehyde is: (a) $SnCl_2 + HCl$ (b) $H_2, Pd/BaSO_4$ (c) $LiAlH_4$ (d) $Zn - Hg/conc.\ HCl$ * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

Q9. Which of the following methylamines is the **strongest base** in an aqueous solution? (a) NH_3 (b) CH_3NH_2 (c) $(CH_3)_2NH$ (d) $(CH_3)_3N$ * **Chapter:** Amines | **Confidence:** Very

High

Q10. Which of the following carbohydrates is a non-reducing sugar? (a) Glucose (b) Fructose (c) Maltose (d) Sucrose * **Chapter:** Biomolecules | **Confidence:** Very High

Q11. The industrial preparation of phenol by heating chlorobenzene with aqueous NaOH at 623 K and 300 atm pressure, followed by acidification, is known as: (a) Dow's Process (b) Cumene Process (c) Raschig Process (d) Kolbe's Process * **Chapter:** Haloalkanes and Haloarenes / Alcohols | **Confidence:** Very High

Q12. What type of isomerism is specifically exhibited by the complex $[Co(en)_3]Cl_3$? (a) Geometrical isomerism (b) Optical isomerism (c) Linkage isomerism (d) Ionization isomerism * **Chapter:** Coordination Compounds | **Confidence:** High

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d). *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q13. Assertion (A): Glycine is the only optically inactive α-amino acid. **Reason (R):** Glycine does not contain any asymmetric (chiral) carbon atom. * **Chapter:** Biomolecules | **Confidence:** Very High

Q14. Assertion (A): The rate of a chemical reaction dramatically increases with an increase in temperature. **Reason (R):** An increase in temperature heavily increases the fraction of reactant molecules possessing energy equal to or greater than the activation energy, leading to more effective collisions. * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q15. Assertion (A): Benzaldehyde does not undergo Aldol condensation when treated with dilute alkali. **Reason (R):** Benzaldehyde does not possess an alpha (α) hydrogen atom. * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

Q16. Assertion (A): Transition metals act as excellent catalysts in many industrial chemical reactions. **Reason (R):** Transition metals exhibit variable oxidation states and can form unstable intermediate compounds with reactants. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

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SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. A first-order reaction has a rate constant $k = 5.5 \times 10^{-14}\ s^{-1}$. Calculate the half-life ($t_{1/2}$) of this reaction. * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q18. State Kohlrausch's law of independent migration of ions. Write the specific mathematical expression for the limiting molar conductivity (Λ_m°) of Barium Chloride ($BaCl_2$). * **Chapter:** Electrochemistry | **Confidence:** Very High

Q19. Write the structure of the distinct organic products formed when aniline reacts with: (i) Bromine water at room temperature. (ii) Acetic anhydride in the presence of pyridine. * **Chapter:** Amines | **Confidence:** High

Q20. Give chemical reasons for the following observations: (i) Zinc (Zn), Cadmium (Cd), and Mercury (Hg) are generally not considered true transition metals. (ii) Scandium ($Z = 21$) is a transition element but does not exhibit variable oxidation states. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q21. Explain the nature of bonding in metal carbonyls. What is meant by *synergic bonding* (pi-backbonding) and how does it strengthen the metal-carbon bond? * **Chapter:** Coordination Compounds | **Confidence:** High

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SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. A 1.00 molal (m) aqueous solution of a weak acid HX is found to have a freezing point of $-1.93°C$. Calculate the degree of dissociation (α) of the weak acid in this solution. *(Given: K_f for water =* $1.86\, K\, kg\, mol^{-1}$*)* * **Chapter:** Solutions | **Confidence:** High

Q23. Answer the following reasoning-based questions: (i) Why is an S_N1 reaction always accompanied by racemization when carried out on an optically active alkyl halide? (ii) Identify the chiral molecule out of 1-chlorobutane and 2-chlorobutane. (iii) Why are haloarenes extremely less reactive towards nucleophilic substitution reactions compared to haloalkanes? (State one primary reason). * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q24. Give the structures and IUPAC names of the main organic products formed when Phenol reacts with: (i) Concentrated HNO_3 in the presence of concentrated H_2SO_4. (ii) Bromine (Br_2) in Carbon disulphide (CS_2) at 273 K. (iii) Chloroform ($CHCl_3$) in the presence of aqueous NaOH followed by acidification. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q25. Give simple and specific chemical tests with observable outcomes to distinguish between the following pairs of compounds: (i) Benzoic acid and Ethyl benzoate (ii) Benzaldehyde and Acetophenone (iii) Ethanal and Propanone * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q26. Write the complete and balanced ionic equations for the following important redox reactions: (i) Oxidation of Fe^{2+} by dichromate ions ($Cr_2O_7^{2-}$) in an acidic medium. (ii) Oxidation of iodide ions (I^-) by permanganate ions (MnO_4^-) in a neutral or faintly alkaline medium. (iii) The action of heat on solid Potassium permanganate ($KMnO_4$). * **Chapter:** d- and f-Block Elements | **Confidence:** High

Q27. Answer the following biochemical questions: (i) Differentiate between globular and fibrous proteins. Give one classic example of each. (ii) What is the exact effect of denaturation on the primary, secondary, and tertiary structures of a protein? (iii) Name the specific water-soluble vitamin whose deficiency causes the disease Scurvy. * **Chapter:** Biomolecules | **Confidence:** Very High

Q28. The rate constant of a chemical reaction is $1.2 \times 10^{-3}\, s^{-1}$ at $300\, K$ and $2.4 \times 10^{-3}\, s^{-1}$ at $310\, K$. Calculate the activation energy (E_a) of the reaction. *(Given:*

$R = 8.314\, J\, K^{-1}\, mol^{-1}$, $\log 2 = 0.3010$) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

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SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Ideal and Non-Ideal Solutions According to Raoult's law, for a solution of volatile liquids, the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution. Solutions that obey Raoult's law strictly over the entire range of concentration are known as ideal solutions. However, many liquid mixtures do not obey Raoult's law and are called non-ideal solutions. These exhibit either positive or negative deviations from Raoult's law. In some cases, liquid mixtures form azeotropes, which are binary mixtures having the same composition in the liquid and vapour phases and boiling at a constant temperature. **(i)** Define the term 'azeotrope'. (1 Mark) **(ii)** What specific type of deviation from Raoult's law is shown by a mixture of ethanol and acetone? Give a brief molecular reason. (1 Mark) **(iii)** State the signs of the enthalpy of mixing (ΔH_{mix}) and the volume of mixing (ΔV_{mix}) for a non-ideal solution exhibiting a *negative* deviation from Raoult's law. Provide one example of such a mixture. (2 Marks) * **Chapter:** Solutions | **Confidence:** High

Q30. Case Study 2: Carbohydrates and Glycosidic Linkages Carbohydrates are primarily produced by plants and form a very large group of naturally occurring organic compounds. They are chemically defined as optically active polyhydroxy aldehydes or ketones. Monosaccharides are the simplest carbohydrates and cannot be hydrolysed further. When two monosaccharide units are joined together with the elimination of a water molecule, they form a disaccharide. The oxide linkage connecting the two monosaccharide units is called a glycosidic linkage. Depending on whether the reducing carbonyl groups are free or involved in bond formation, carbohydrates are classified as reducing or non-reducing sugars. **(i)** Name the two specific monosaccharide units obtained upon the complete acid hydrolysis of Sucrose. (1 Mark) **(ii)** What is the specific type of glycosidic linkage present in Maltose? (1 Mark) **(iii)** Explain structurally why sucrose is a non-reducing sugar, whereas maltose is considered a reducing sugar. (2 Marks) * **Chapter:** Biomolecules | **Confidence:** Very High

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SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Electrochemistry: Theories and Thermodynamics (i) Define limiting molar conductivity (Λ_m°). Explain why the limiting molar conductivity of weak electrolytes (like acetic acid) cannot be determined experimentally by extrapolating the plot of Λ_m versus $\sqrt{c}$ to zero concentration. (2 Marks) (ii) The standard electrode potential (E°_{cell}) for the Daniell cell is $1.10\, V$. Calculate the standard Gibbs free energy change (ΔG°) and the equilibrium constant (K_c) for the cell reaction: $Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ *(Given:* $1F = 96500\, C\, mol^{-1}$, $R = 8.314\, J\, K^{-1}\, mol^{-1}$, $T = 298\, K$*)*. (3 Marks) * **OR (B)** (i) Describe the construction of a common Dry Cell (Leclanché cell). Write the specific chemical reactions occurring at the zinc anode and the graphite cathode during its discharging. (2 Marks) (ii) An aqueous solution of Copper(II) sulphate ($CuSO_4$) is electrolysed for 10 minutes using a steady current of 1.5 Amperes. What is the mass of copper deposited at the cathode? *(Given: Molar mass of Cu =* $63.5\, g\, mol^{-1}$*,* $1F = 96500\, C\, mol^{-1}$*)*. (3 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q32. (A) Coordination Compounds: VBT and Isomerism (i) Using Valence Bond Theory (VBT), deduce the hybridization, geometry, and magnetic property (paramagnetic or diamagnetic) of the complex ion $[NiCl_4]^{2-}$. *(Given: Atomic number of Ni = 28, and chloride is a weak field ligand)*. (3 Marks) (ii) What is meant by crystal field splitting energy (Δ_o)? How does the magnitude of Δ_o relative to the pairing energy (P) determine whether a d^4 metal ion will form a high-spin or low-spin complex in an octahedral field? (2 Marks) * **OR (B)** (i) State any two main postulates of Werner's theory of coordination compounds. How did Werner explain primary and secondary valencies? (2 Marks) (ii) Draw the crystal field splitting diagram for d-orbitals in an **octahedral** crystal field, clearly labelling the t_{2g} and e_g energy levels. (2 Marks) (iii) Write the systematic IUPAC name for the coordination complex: $[Pt(NH_3)_2Cl(NO_2)]$. (1 Mark) * **Chapter:** Coordination Compounds | **Confidence:** Very High

Q33. (A) Organic Chemistry: Name Reactions Provide the balanced chemical equations and the specific reaction conditions for the following important name reactions: (i) Swarts reaction (ii) Fittig reaction (iii) Kolbe’s reaction (iv) Wolff-Kishner reduction (v) Hofmann bromamide degradation (5 Marks) * **OR (B)** An organic compound **(A)** with the molecular formula $C_5H_{10}O$ gives a positive 2,4-DNP test but a negative Tollens' test. It also does **not** give a yellow precipitate with iodine and aqueous NaOH (negative Iodoform test). Upon reduction with Zinc amalgam and concentrated HCl, it produces a straight-chain hydrocarbon **(B)**. When compound **(A)** undergoes drastic oxidation with hot acidic $KMnO_4$, it yields a mixture of two distinct carboxylic acids, **(C)** and **(D)**. (i) Deduce the structures of compounds **(A)**, **(B)**, **(C)**, and **(D)** and write their IUPAC names. (4 Marks) (ii) Name the specific reduction reaction used to convert compound **(A)** to hydrocarbon **(B)**. (1 Mark) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids / Haloalkanes | **Confidence:** Very High

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